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Optimization of process parameters of catalytic pyrolysis using natural zeolite and synthetic zeolites on yield of plastic oil through response surface methodology

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This study aims to reach a sustainable solution for waste management of medical plastics through value-added product extraction. It uses the DOE technique to examine the effect of natural zeolite and synthetic Al_2O_3 and SiO_2 as catalysts. A small lab-scale pyrolysis setup was used for medical plastic waste management treatment. Pyrolysis of medical plastics with temperature range (350–450 °C), three catalysts, and wt.% are examined. This process is designed for 3 factors and 3 levels, such as type of catalyst, catalyst wt.%, and temperature, to create an L9 orthogonal array. At the same time, the heating rate and residence time are maintained constant at 5 °C/min and 75 minutes, respectively. Furthermore, this study analyzed the input variables in catalytic pyrolysis using response surface methodology. As a result of the study, generating the regression equation for oil yield, F and P values assure the model is significant. Optimization result shows the type of catalyst, temperature, and catalyst concentration values are found as aluminum oxide, 376 °C, and 6.6 wt.%, respectively. HDPE and LDPE oil yield a value of 58.3648 and 61.2051 wt%, respectively, under the optimum variables condition. For oil yield prediction, HDPE and LDPE's correlation coefficient (R^2) were 0.9949 and 0.9943, respectively. Authentication of the model response using a regression equation validated with the experimental result shows good agreement. The produced medical plastic oil has a density, viscosity, flash & fire point, carbon residue, and cetane number 904 kg/m^3, 2.3 cSt, 42 & 45 °C, 7.1 wt.% and 51 respectively. Finally, this study concludes that plastic oil extraction from medical waste through catalytic pyrolysis can be a potential source of alternative fuels in IC engines. Priority to optimization and low-cost catalysts highlights medical plastics waste management under the socio-economic model.

Keywords Plastic oil, Optimization, Pyrolysis, RSM

In the decade, catalytic pyrolysis has been given attention in the energy recovery and waste management sector[1]. Waste management is a significant challenge for every country, especially hazardous wastes like plastics, which require several years to decompose. Almost all industrial entities in the modern world depend on plastic materials as they are adapted to various design criteria and have led to technological advancements. Unfortunately, this has resulted in a substantial accumulation of waste plastics, which is rising at an alarming rate[2]. As of 2021, plastic production has reached 380 million tons per year, and the total production of plastic has reached 8 billion tons.

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This means that there is more than one ton of plastic for every person alive today. The absolute change in plastic production between 1950 and the past decade was estimated as 379 million tons per year, with a relative change of more than 18.59%. In India, more than 15 million pieces of plastic are produced yearly, yet proposed waste management and decomposition remains a significant challenge and requires a long-term and viable solution[3].

Plastic has played a significant role in modernizing various industries, particularly healthcare[4]. Healthcare equipment uses significant plastics due to their durability, cost-effectiveness, etc[5]. However, plastic waste from medical equipment is a serious concern as it might contain public health risks. For example, 8 billion vaccine doses and 140 million test kits were generated for COVID-19, and most of them contained plastics. An overview of the Indian medical plastic market is shown in Fig. 1. Results show a steady increase in the use of plastics for medical purposes. Therefore, the WHO has requested the reform of medical plastics due to their abundant impact on the environment and on health. The current solution to hazardous medical plastics uses either landfills or incineration. However, incineration causes the production of toxic and greenhouse gases while landfills take up empty land, deeming it inhabitable due to the presence of waste[6].

The production of plastics requires the use of crude oil and its derivatives. As most countries do not produce enough oil to sustain national plastic production, plastic production depends on imports and price fluctuations in the international market. To address this issue, many researchers have recommended using biodiesel from different feedstock as an alternative[7]. However, biodiesel utilization has some downsides, such as the availability of land, the oil's stability, the extraction process's complexity, and seasonal feedstock[8].

To address the issue of crude oil imports and plastic waste management, it is important to realize that plastic waste contains hydrocarbons. Any attempt to extract the hydrocarbons from waste plastics would solve the issue of waste management and reliance on oil imports. A source of energy recovery from plastic through pyrolysis is a sustainable solution to plastic waste management. Pyrolysis is a thermo-chemical process that occurs in a closed environment. Compared to other pyrolysis processes, incineration pyrolysis is an environment-friendly waste management approach. Czajczyńska et al.[9] studied the potential of energy recovery from pyrolysis using organic and inorganic materials. Qureshi et al.[10] also studied the possibilities and challenges of plastic waste management by pyrolysis. Due to socioeconomic benefits, they concluded that pyrolysis provides a great alternative to incineration and landfilling. Pyrolysis has also been successfully used for waste management of medical applications. Velghe et al.[11] experimented with oil production from municipal solid waste and identified the distillation temperature as the most critical parameter. Ding et al.[12] studied the pyrolysis dynamics of syringes and medical bottles as feedstock. They were able to produce aromatic compounds up to C24 and C41. The present study selects high-density and low-density polyethylene feedstock for investigation.

Selection of feedstock and catalyst

Due to their adaptability, new inventions and revolutions of material compounds highly depend on plastic. Therefore, the demand for and consumption of plastics is increasing perpetually. For example, in the Indian market, Fig. 2a&b elucidates plastic consumption and demand, including which type of plastic is used. Polyethylene (PE) has the most significant demand share of all plastics, accounting for 33% of the total market, and consumed over 4600 kilotons annually from 2019 onwards[13]. Plastic accumulation keeps rising due to the need for effective plastic waste management. Several researchers have proposed various strategies and methods for plastic waste management; however, this was still transient. In common, PE plastics were high-density polyethylene (HDPE)

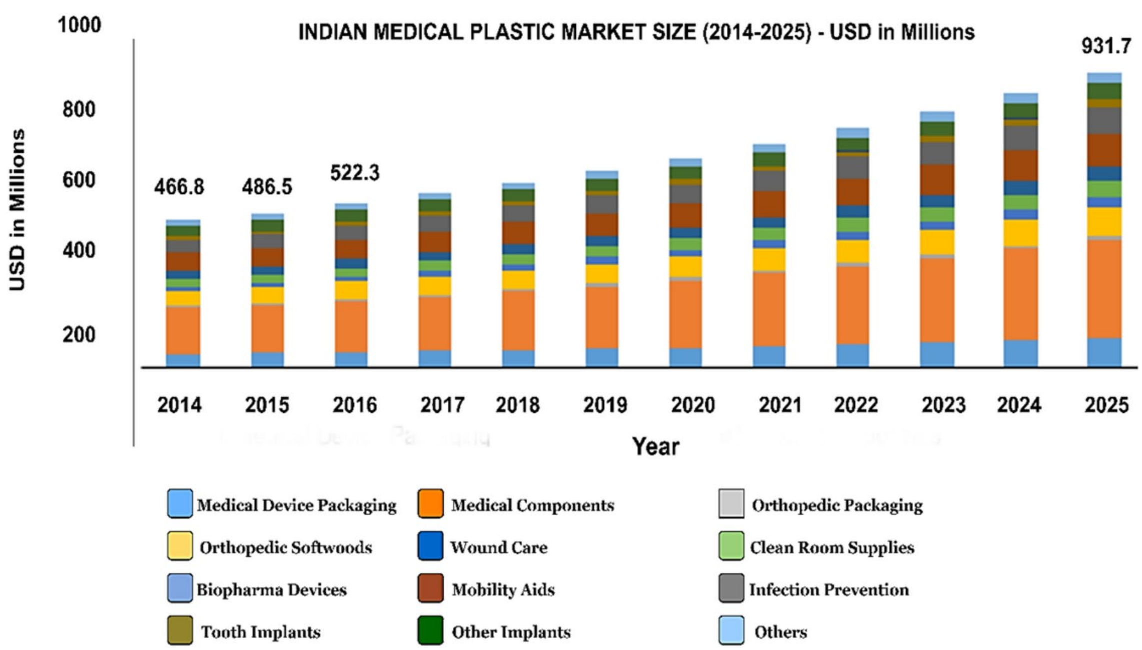

Fig. 1. Indian plastic market size (by application) - 2014–2025[4].

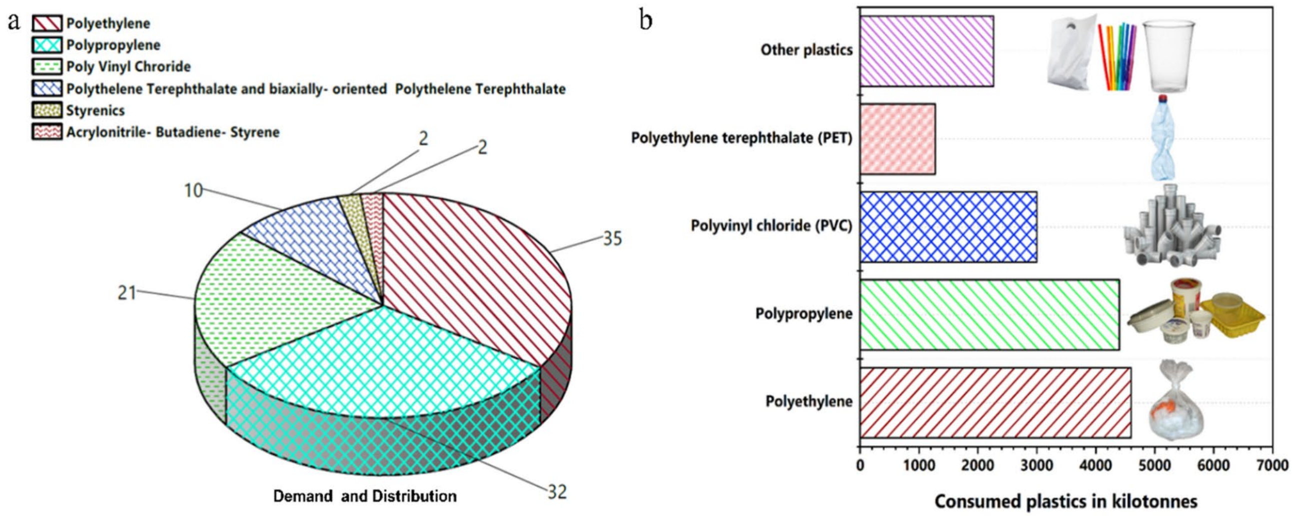

Fig. 2. Distribution of plastic-type and their demand in the Indian market - 2019 onwards.

and low-density polyethylene (LDPE). Based on the available statistics on PE plastics, HDPE and LDPE were selected for the present study. Catalyst selection is based on consistent previous literature, and zeolite catalysts are before other catalysts for plastic pyrolysis[14]. The vital reason behind this is that zeolite-based catalysts possess high surface area, hydrothermal stability, bronsted solid acid sites, and higher structural pores of molecular dimensions[15]. Furthermore, zeolite-based catalysts were chosen for this study due to their high acid value, a superior property for plastic thermal degradation[16]]. Catalysts can substantially reduce the activation energy and support oil extraction using catalysts. Miandad et al.[17] showed more than 30% improvement in yield and a better oil grade. In addition, energy consumption was reduced. A similar study by Kurniawati et al.[18] showed an improvement of more than 24.5 by wt. % with a reduction in char. Similarly, Shah et al.[19] investigated an acid catalyst and concluded that it yielded a higher yield than a base catalyst. Maqsood et al.[20] conducted a comprehensive study about plastic pyrolysis and discussed some catalyst results from past works amongst the various types of catalysts used in literature. Zeolites are one of the commonly used catalysts. A high yield was achieved using pyrolysis and a modified natural zeolite catalyst. Susastriawan et al.[21] carried out pyrolysis of LDPE with various sizes of zeolite and showed yield improvement based on various sizes. Rahimi et al.[22]] used a high siliceous ZSM-5 nanocatalyst and obtained hydrocarbons in the C_6 - $C_1 2$ range. These studies conclude the importance of catalysts in pyrolysis. In addition, catalyst application was most helpful for recovering oil from waste plastics. As mentioned, optimizing the pyrolysis process is important to extract the maximum yield. This work investigates the behavior using natural zeolite, alumina, and silica catalysts. Specifically, this paper investigates a comparison of both feedstocks (HDPE & LDPE) and their behavior along with oil yield concerning the low-cost catalyst, low catalyst concentration, and minimum temperature.

Importance of process optimization

The operational behavior of pyrolysis and its end product depend on various parameters. Hence, effective tuning of this parameter is essential to attain higher production. Like catalysts, process optimization can also be used to improve the process efficiency of pyrolysis. This makes it possible to find a way to improve oil yield productivity, effective resource utilization, and energy consumption. Process parameters such as temperature, pressure, residence time, reactor type, bed material, heating rate, and quantity of the feedstock affect the pyrolysis process[23]. Furthermore, the end product requires specific agendas, particularly for oil yield, which is crucial to tune parameters. Which was fall to tune, gas products are increasing. Response surface methodology (RSM) is an emerging trend in various process optimization. More information is needed from a minimum number of experiments. RSM, generate models, fit detail summary, and main effects of input variables and their correlation on the response. Jha et al.[24] used DOE to find the optimal catalyst ratio, temperature, and residence time. Results from consistent literature show the importance of DOE in optimizing process parameters to extract the maximum yield from the pyrolysis process. However, more work is needed to maximize the use of statistical approaches for optimizing the pyrolysis process. Therefore, based on the previous consistent literature, predicting the optimal region is crucial to oil extraction among gas and char yield. Moreover, the interaction effect of zeolite-based catalysts and their concentration relative to temperature is rarely validated. In the present study, considering the disposal of plastic medical waste is a significant concern, pyrolysis can be used as a viable solution. This work uses synthetic and natural zeolite catalysts to perform pyrolysis on medical waste. In addition, a DOE approach is also used to optimize the process based on the type of catalyst, temperature, and concentration of the catalyst. The objective of the optimization performed in this work was to reduce the energy input and increase the process yield to enable the adoption of catalytic pyrolysis as a large-scale industrial process.

Materials and methods

Experimental setup

The medical plastic waste, such as full-sleeve gloves, masks, syringes, and shoe covers, was collected from the nearest healthcare hospitals with proper safety precautions. The collected plastic waste was preheated by sterilizing it to a higher temperature of 121 °C. In this work, high-density polyethylene (HDPE) and low-density polyethylene (LDPE) plastics were used after being shredded, washed, and dried A small lab-scale bench-type pyrolysis reactor with a 2kW electric heater was used in this work (Fig. 3). Waste plastics were treated at a temperature of 350, 400, and 450 °C with a heating rate of 5 °C/min and a starting capacity of 2000 ml with maximum weight of 1500 g. The gases were condensed to liquid oil using a water-cooled double-column boron tube condenser at 27–30 °C. Three catalysts, namely aluminum oxide, silicon oxide, and natural zeolite, with different concentrations of 4 wt.%, 5.3 wt.%, and 6.6 wt.%, respectively, were used for experimentation. A total of 18 experiments are carried out, which means 9 experiments are conducted (3+3+3) on each plastic, for example, alumina catalyst with a temperature range of 350, 400, 450 °C and catalyst concentration (4, 5.3, 6.6 wt%). Simultaneously, silica and natural zeolite were used on each plastic.

$$\text{Oil yield(wt.\%)} = \frac{\text{Weight of oil produced}}{\text{initial Sample weight-weight of spent shale}} \tag{1}$$

Design of experiment (DOE) statistical analysis

Response surface methodology modeled and optimized the response according to the factors The RSM featured for (a) design of experiments with a reliable response, (b) developing a best fit mathematical model from DOE, (c) finding the optimal value under the required optimization criteria shown in Table 2 and experimental design matrix. The present study was executed by face central composite design (FCCD), which consists of 8 cube points, 6 center points on the cube, and 6 axial points, therefore resulting in a total of 20 experiments. These experiments are performed randomly. Performed experimental results were fitted to a suitable model like a quadratic polynomial model, and predicted values are solved by this derived equation for both HDPE oil yield and LDPE oil yield. These results were graphically represented through contour plots. In the present study, optimization is performed based on a design of experiment (DOE) approach. The response surface methodology experimental design is one of the most popular statistical methods and has been used to optimize various problems[25]. An L9 orthogonal array was used to study the parameter space. In this work, the Type of Catalyst, Catalyst wt%, and temperature are input process parameters. HDPE and LDPE plastics oil yield are selected as an output response. The results from the design are used to optimize the pyrolysis process, resulting in maximum yield with energy conservation and effective resource utilization. The desired range of parameters and goals set in the optimization is mentioned in Table 1.

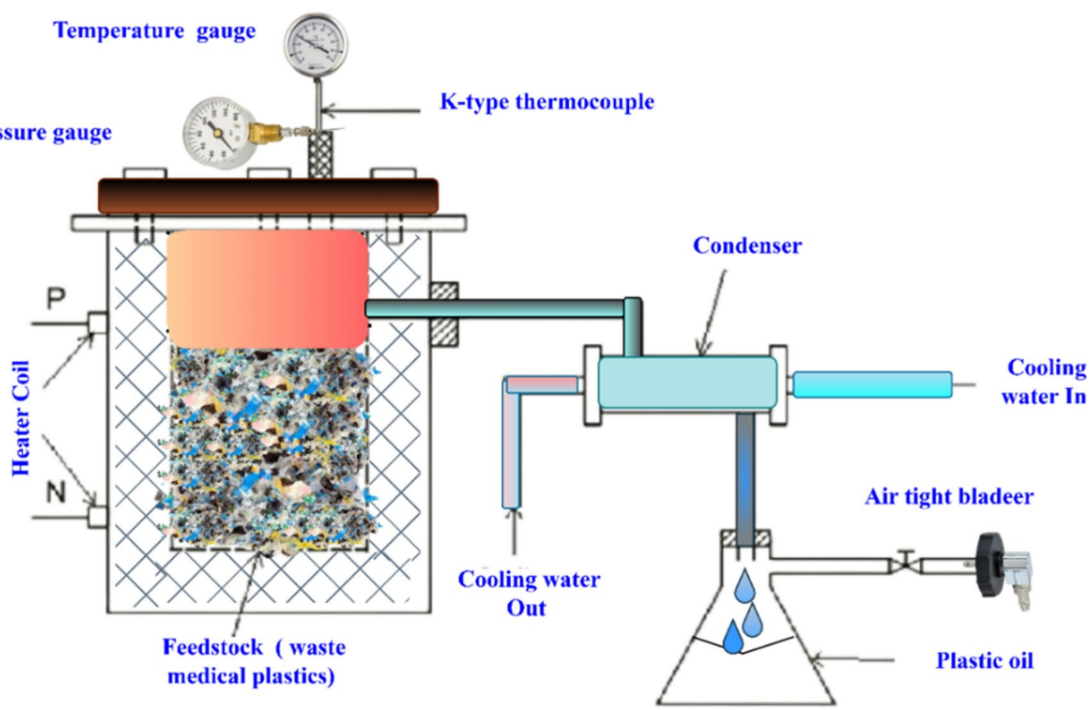

Fig. 3. Small lab-scale pyrolysis- setup.

Criteria	Goal	Limits
Temperature °C	Minimise	400 °C
Catalyst type	In range	0–3
Catalyst wt%	In range	0–6.6
HDPE oil yield	Maximum	100%
LDPE oil yield	Maximum	100%

Table 1. Optimisation criteria.

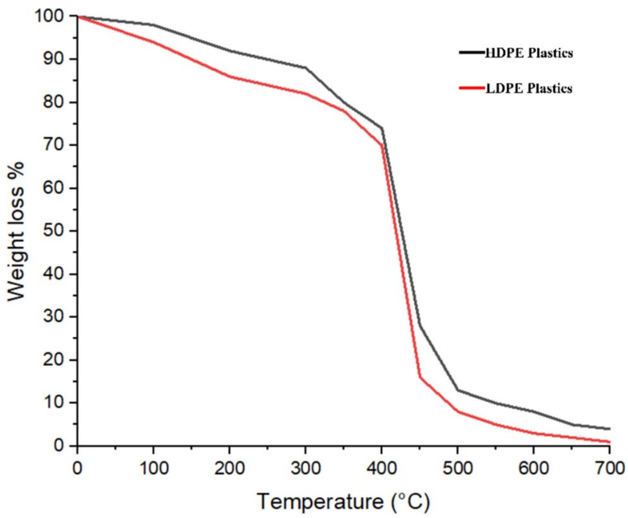

Fig. 4. Thermogravimetric analysis.

Results and discussion
Thermogravimetric analysis
Thermogravimetric analysis (TGA) analysis is used to predict the optimal conversion temperature range. This helps in finding the thermal stability of the feedstock medical plastics and the material behavior in an anti-oxidizing environment (Fig. 4). The TGA technique determines the highest conversion region of HDPE and LDPE plastics to execute the process. At 400 °C, solid was present up to 33.05%, but at a short period, the amount is reduced to 0.54% within 500 °C. The maximum conversion is obtained from HDPE at 420 °C. The predominantly liquid yield produced from the conversion is 80.88 wt.%. In another sample, LDPE normally has weaker intermolecular force, tensile strength, and hardness compared to HDPE, whereas the ductility of the LDPE is better than HDPE. Conversation of these medical plastics highly occurs between 420 °C with more form of liquid 85.68%. Maximum liquid conversions occur between 380 and 500 °C in both cases. It is found that 420 °C is the optimized process temperature for executing the pyrolysis process for the feedstock LDPE and HDPE medical plastics used in this work by TGA. However, it is possible to reduce the temperature using a catalyst. This study found the optimal temperature by experimenting with a catalyst and using RSM relative to maximum oil yield.

Process parameters and its influence on oil yield
RSM plots in Figs. 5, 6 and 7 show the corresponding results from the statistical analysis of the proposed Taguchi design. In general, results show that temperature plays an important role in the yield of the pyrolysis process. A high liquid yield of 95 wt.% is obtained with low gas yield and negligible char at 420 °C. On the other hand, the temperature range above 420 °C increases the gas yield from both HDPE and LDPE. In addition, both plastic samples produced better yields with all three catalysts, and an increase in catalyst ratio increased the yield and reduced the energy consumption.

Effect of type of catalyst
Thermal pyrolysis is highly energy intensive due to low thermal conductivity and endothermic cracking; therefore, a catalyst is used to overcome this issue. Each catalyst has signature properties and a diverse impact on the pyrolysis process and end products. Natural zeolite, alumina, and silica catalysts are selected in the study because they are available more at a low cost, and most importantly, previous studies have proposed zeolite-based catalysts for plastic pyrolysis due to their high acid value[14]. The effect of the catalyst on the pyrolysis liquid oil yield concerning various temperatures and concentrations is shown in Fig. 5a–d. Natural zeolite catalysts' unique pore topology, textural properties, and Bronsted and Lewis acid sites synchronize the feedstock molecules for effective reaction results; a high rate of C-C reaction accelerates thermal degradation[26]. Similarly, the type

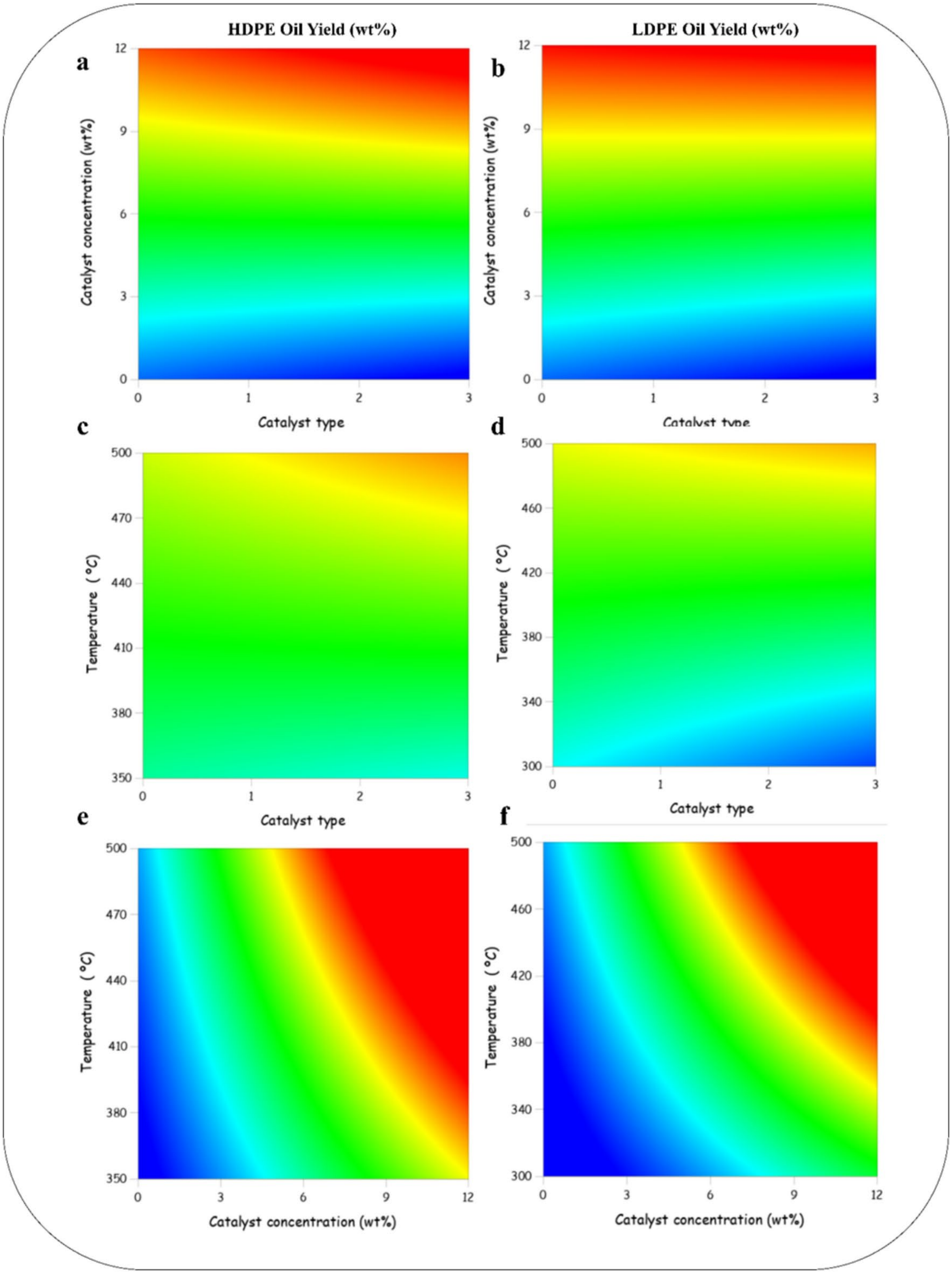

Fig. 5. Contour Plots of HDPE and LDPE Plastic Oil Yield wt.% (**a**, **b**) Catalyst concentration, (**c**, **d**) catalyst type, and (**e**, **f**) reaction temperature.

of catalyst also affects the yield as every catalyst has a different thermal conductivity, heat dissipation rate, morphology, acid value, etc. The present study shows a similar trend at the starting stage compared to alumina catalyst natural zeolite recorded 2-3 wt.% more liquid yield at 400 °C and 5.3 wt%. However, further increases in temperature and catalyst concentration sequentially decrease heavy fractions and increase light olefin formation, promoting gas production; therefore, cumulative liquid production from natural zeolite catalyst is low against alumina by up to 20%. Natural zeolite catalyst produced a lower liquid yield of plastic oil than aluminum oxide

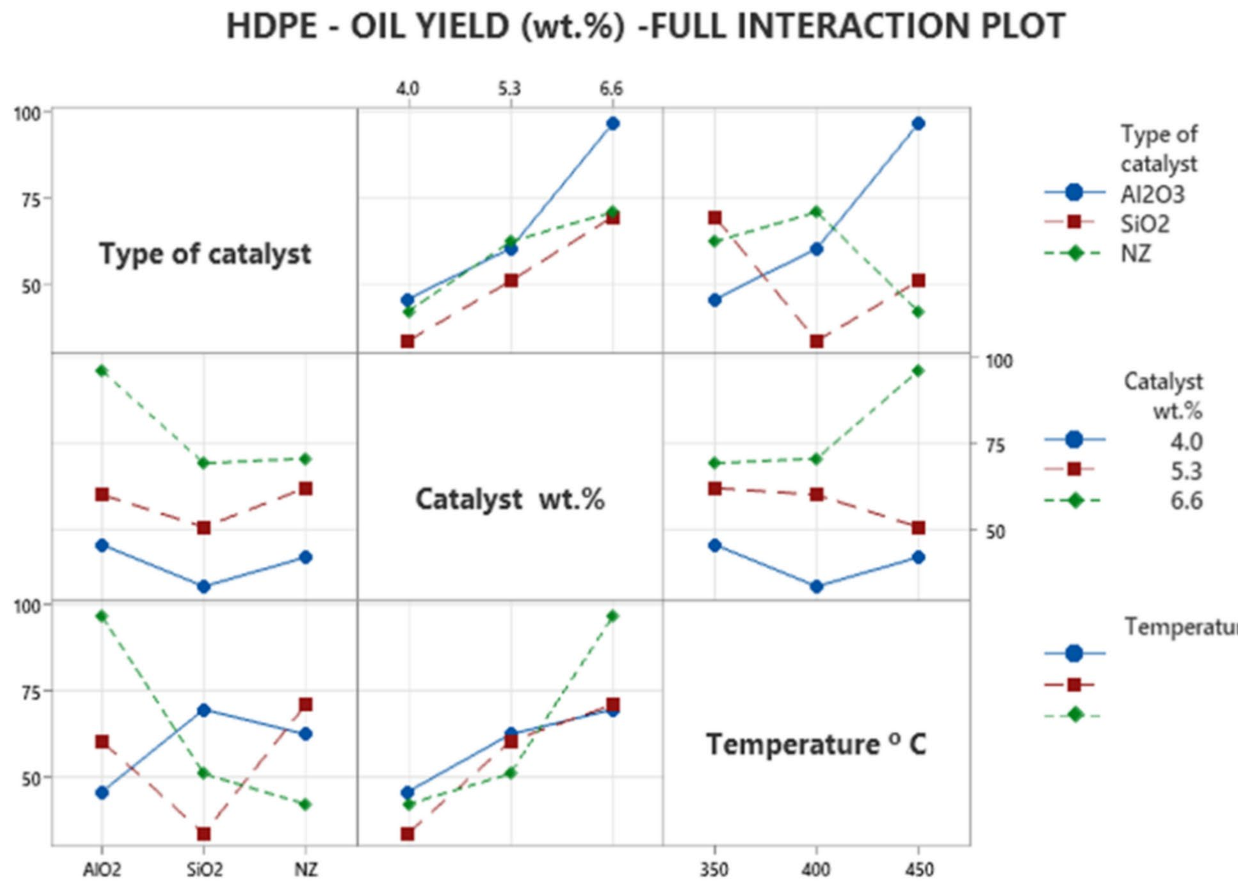

Fig. 6. Full interaction plot for HDPE oil yield (wt.%).

at 450 °C. Alumina has high chemical inertness and possesses a tremendous surface-to-volume ratio. Therefore, it extracted more yield than natural zeolite and silica, with 20% and 40% values, respectively. In addition, the pore diameter is a major influence on thermal degradation; the pore diameter of alumina is around 8 nm[27], which was helpful prior to the thermal decomposition of feedstock against the other two catalysts. Therefore, the contrast is reflected in the oil yield. Fig. 5c,d explores respective oil yields with correlation to temperature and type of catalyst. On the x-axis, 1, 2, and 3 correspond to alumina, natural zeolite, and silica catalysts. At 350, 400, and 450, oil yield by alumina and silica catalyst was recorded as 45.6, 60.2, 96.4, and 42, 62.2, and 70.8 wt%, respectively, from HDPE plastics. On the other hand, LDPE had 47.6, 63.6, 98.3, and 44, 63.9, and 72.9 wt%, respectively. Natural zeolite catalyst enhanced gas yield from 400 °C. The 20% net increase in oil yield is attended by an aluminum oxide catalyst against natural zeolite, and silica performs vice versa to alumina. These are due to diversity at the molecular level, such as pore size, structure and crystallinity, and acid value.

Effect of catalyst concentration

The effect of catalyst concentration on oil yield (4.5 wt% to 6.6 wt%) concerning various temperatures and catalysts, as shown in Fig. 5a, b, e, and f, typically, an increase in catalyst concentration increases more active contact between feedstock. Multitudinous researchers have reported up to 3 to 12 wt% catalyst concentration as the suitability of oil yield. Gas yield increases with the increase of catalyst[28]. Maximum oil yields are obtained at a catalyst concentration of 6.6 wt.% to all temperature ranges and both plastics. Based on hefty diversity consistent with previous literature, feedstock and catalyst ratios up to 12 wt.% catalysts support oil yield better. More augmented catalysts in the feedstock ratio lead to more gas products[29]. Figure 5a&b displays the oil yield based on the type of catalyst and catalyst concentration. All catalysts attained optimized yield at 6.6 wt.% concentration. Further concentration increases enhanced the gas yield and reduced the oil yield by 5.3 wt.% concentration of NZ, alumina, and silica extracted 62.2, 60.2, and 50.8 wt% oil yields, respectively, from HDPE and 63.9, 63.6, and 52.6 wt.% from LDPE, respectively. The results clearly show the importance of catalyst concentration as it influences the oil yield.

Effect of reaction temperature

Typically, an increase in temperature increases the rate of cracking. In pyrolysis, consistent with literature from previous studies, an increase in temperature increases oil yield up to critical temperature; the critical temperature varies by feedstock and operational behavior. High temperatures quickly break the chains of molecules in the feedstock so that volatile compounds rapidly evaporate sequentially to augment non-condensable gas's end

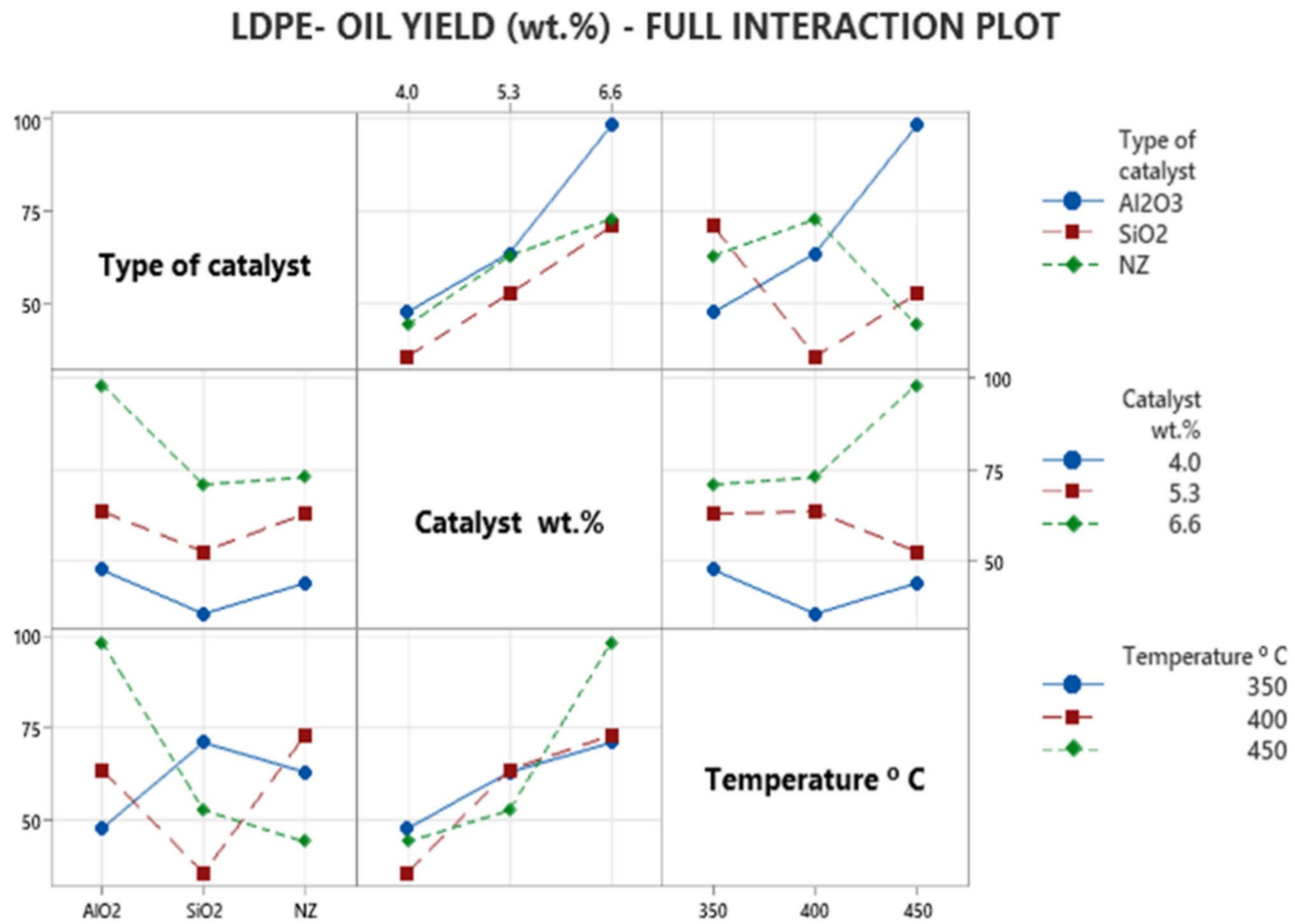

Fig. 7. Full interaction plot for LDPE oil yield (wt.%).

product[30]. The pyrolysis process and end products are radically different by the reaction temperature. An increase in temperature increases thermal degradation, mainly above 500 °C, and more light olefins are formed, indicating the augmentation of the gas product[31]. The oil yield rate is high at the range of 350 to 450 °C. The present study restricted the temperature range to 450 °C. The temperature and catalyst concentration correlation for HDPE and LDPE plastic oil yield was exhibited in 2D plots in Fig. 5e&f. It is well known that temperature and catalyst concentration are directly proportional to oil extraction because an increase in temperature and the presence of a catalyst accelerates thermal degradation[32]. Results show that increased temperature and catalyst concentration increase the oil yield. However, after 420 °C, the rate of oil yield extraction decreased as gas formation increased. This is due to higher temperatures giving kinetic energy to the feedstock particles, and the key to moving based on Charles's law was already proven.

Effect of feedstock HDPE and LDPE to oil yield

The phenomenon can be elucidated as HDPE and LDPE are from the same family, PE. However, they have some unique features. Notably, HDPE is tightly packed together against LDPE due to crystallization differences. HDPE is an opaque linear structure, and LDPE is a transparent branched version of PE. HDPE's melting point is 20 °C higher than LDPE. LDPE melting point was 115 °C[33]. Meticulously found, this was the reason for HDPE and LDPE oil yield contrast. In TGA analysis, HDPE and LDPE exhibit a similar thermograph with a single mass-loss zone with a difference of around 10–20 °C. The contours and surface plots for the effect of catalyst concentration and catalyst type on HDPE and LDPE plastic oil yield are shown in Fig. 5a and b, respectively. Results showed that the LDPE plastic type obtained a higher oil yield at all stages of process conditions. For both HDPE and LDPE feedstock, an increase in temperature and catalyst concentration enhances the oil yield. Up to 380 °C, the oil yield difference between LDPE and HDPE was observed to be around 10 wt.%. Figure 5e,f Further increase of temperature and catalyst concentration improve the HDPE plastic oil yield substantially; consequently, at a maximum region and above 400 °C, both HDPE and LDPE plastic oil extraction contrast are shrinkage to 2-5 wt%. HDPE's bulky molecular size, when compared to LDPE, is thought to be a significant contributing factor to this behavior[34]. It should be noted that catalyst cracking ability depends mainly on the molecular size of the feedstock and catalyst and their relative quantity. This phenomenon is key for plastic oil extraction energy conservation by minimizing the process temperature. It is well known that catalyst usage minimizes the activation energy and improves the oil quality due to a higher level of the fraction[35]. In both cases

Source	Sum of squares	Degree of freedom	Mean square	F	P
Model	1521.55	6	253.41	6.32	0.0362
Type of Catalyst	2.2491	2	1.12	3.94	0.0642
Catalyst wt.%	484.47	2	242.24	5.86	0.0327
Temperature (°C)	1035.26	2	517.96	8.11	0.0137

Table 2. ANOVA Results of HDPE Oil Yield - Model Significant.

Source	Sum of squares	Degree of freedom	Mean square	F	P
Model	1465.53	6	244.25	6.38	0.0362
Type of Catalyst	3.3844	2	1.6922	4.63	0.0642
Catalyst wt.%	468.43	2	234.22	5.286	0.0327
Temperature °C	993.73	2	496.86	9.286	0.0137

Table 3. ANOVA Results of LDPE Oil Yield - Model Significant.

(HDPE and LDPE), the aluminum oxide catalyst provides maximum oil yield over the other two catalysts, NZ and silica oxide. More than 90 wt.% of oil yield was obtained at 380 °C–420 °C and 6.6% Al_2O_3 concentration. Natural zeolite catalyst with 5.3% catalyst concentration and 400 °Cprocess reaction temperature extracted 62.2% and 63.9% oil yield from HDPE and LDPE respectively. At the same condition, oil yield through aluminum oxide extraction was 3 wt.% less because the higher crystallinity of natural zeolite contributed to effective cracking. But above the reaction temperature of 400 °C, natural zeolite catalyst enhanced the gas yield while reducing the oil production. Lastly, silica catalysts with both raw materials showed the least yield in this work. Based on these results, the Al_2O_3 catalyst yielded should be used for maximum yield while a natural zeolite catalyst can be used for a low-energy consumption process with catalyst modification for maximum oil yield based on the investigation input factors selection is crucial to oil yield; therefore, these results show the importance of catalyst modification and making a hybrid catalyst, optimization of concentration, and temperature. This increase in the oil yield, followed by a reduction in the oil yield, was used to find the optimum input factor for oil extraction to achieve maximum yield. This remaining paper tries to predict the optimum parameter within the experimental scheme.

Performance optimization using RSM

In the analysis performed in this work, three input and one output factor were used for the statistical analysis. The three factors are (a) the type of catalyst (Al_2O_3, SiO_2, and NZ), (b) catalyst wt.% (4–6.6 wt.%), and (c) temperature (350–450 °C). The experimental design using the RSM is used to identify the effect of each parameter so that the output can be optimized. Parameter optimization was performed after the experimental design and analysis. 68 iterations were performed. A multi-objective optimization yielded maximum oil yield at the lowest temperature. A low temperature was set as an optimization objective to lower the production costs of pyrolysis. Weight and importance were given equally to all parameters to avoid bias. Optimization results show aluminum oxide with 6.6 wt.% at 376 °C to yield the maximum 5% and 61% yield for HDPE and LDPE, respectively. Tables. 2 and 3 illustrate the fit summary of the designed model. Equality of means (F value) is above 4, and a very minimum lack of fit (P value) assures the model can exhibit the result 95% confidence level at any time and under the same condition. The null hypothesis is discredited.

The regression equation from the model for HDPE and LDPE is given Eqs. 2 and 3.

$$HDPE = 32.5323 - 12.7565 \times A - 3.90194 \times B + 0.0233321 \times C + 0.423077 \times AB + 0.0265 \times AC + 0.0184615 \times BC \quad (2)$$

$$LDPE = 33.9578 - 12.267 \times A - 3.46114 \times B + 0.024796 \times C + 0.269231 \times AB + 0.025 \times AC + 0.0180769 \times BC \quad (3)$$

Figures 6 and 7 visualize the effect of interaction between three factors concerning HDPE and LDPE oil yield, respectively. All lines concerning each factor are not horizontal, indicating three factors mutually affect oil yield. There was a strong three-way interaction between the main effect of temperature, catalyst concentration, and type of catalyst, as evident from the nonparallel nature of the effect lines for the interaction on HDPE and LDPE oil yield. When catalyst concentration is at its low level, temperature strongly affects the oil yield of both LDPE and HDPE oil yield. The interaction of each factor line mentioned the greater strength of the interaction between all factors. The graph also shows that temperature does not interact much with the type of catalyst factor. Catalyst concentration slightly interacts with the type of catalyst factor. All the interactions Al_2O_3 catalyst at a higher level than natural zeolite and SiO_2. Compared to the HDPE and LDPE, interaction plot slope formation is higher in LDPE because HDPE has a longer hydrocarbon chain; hence, the sustainable amount of intermolecular force is required to break the bonds, hence requiring a higher temperature for initial cracking compared to LDPE. By examining the full interaction, the effect of both HDPE and LDPE oil yield depends on

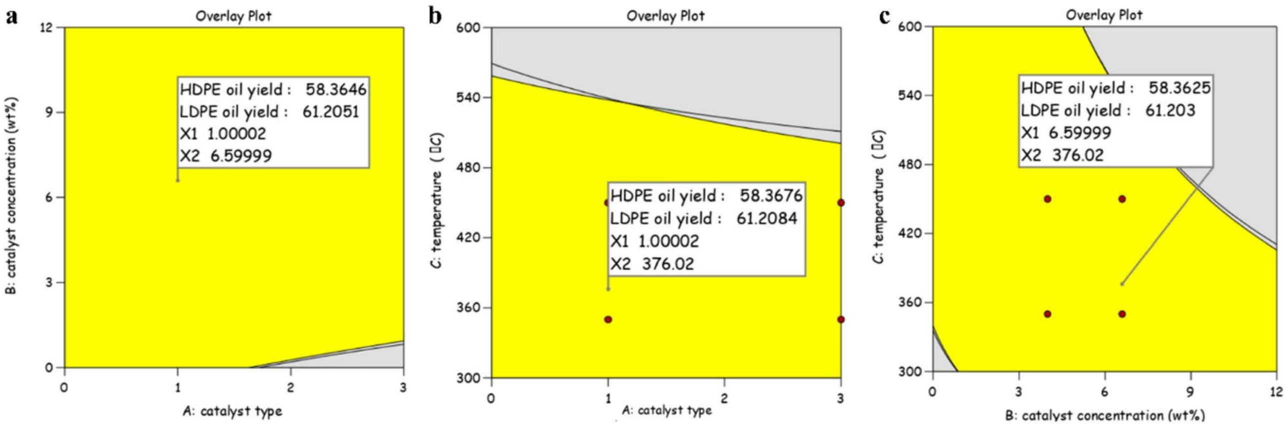

Fig. 8. Overlay Plot (**a**) Catalyst type vs catalyst concentration, (**b**) Catalyst type vs temperature, and (**c**) Catalyst concentration vs temperature.

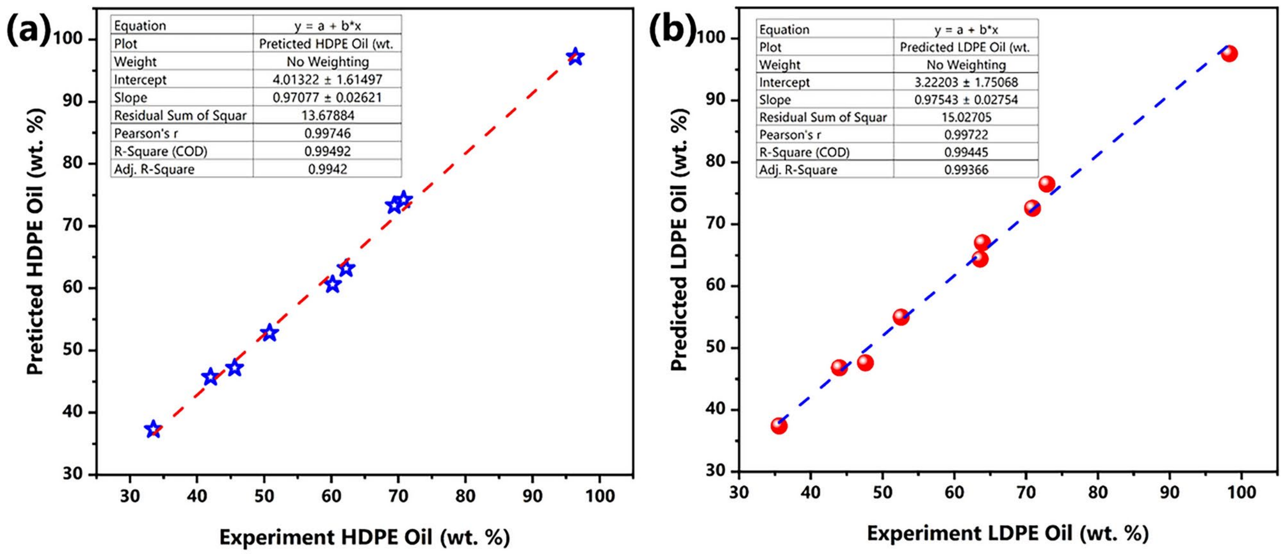

Fig. 9. Linear agreement between experimental yield and predicted yield of (**a**) HDPE and (**b**) LDPE.

the temperature range. The LDPE interaction line has a higher delta value than the corresponding plot line in the HDPE interaction. Figure 8 shows the overlay Plot proposed by the design expert V13 Software, showing design space in yellow relative to optimized formulations with the responses. Optimal response relative to their DoE is pinned and exhibits the response values of 58.355 wt% and 61.1957 wt.% LDPE and HDPE respectively for Al_2O_3, concentration of 6.6 wt% at 376 °C. Results show an improvement in yield with Al_2O_3, a higher catalyst%, and at an optimum temperature. For both HDPE and LDPE oil yield responses, selected factors play a vital role in deciding the oil yield response is almost the same.

ANOVA analysis against the experimental results

The experimentally observed response versus the response predicted by the model is graphically presented in Fig. 9a and b for HDPE and LDPE oil yield, respectively. A reference dotted line is drawn to show a 100% accurate model. For both HDPE and LDPE, the regression coefficient R2 value was found to be 0.9949 and 0.9945, respectively. This high R^2 value shows a near-perfect prediction model, as shown in Fig. 9a and b. Finally, the predicted optimum values are authenticated by the linear regression equation. RSM has the potential to forecast the pyrolysis yield. The prediction of optimization and various interaction investigations are useful for waste plastic catalytic pyrolysis upcycling, especially for oil extraction at optimal process conditions. Extracted oil fuel properties are measured and compared in conventional diesel.

Extracted medical plastic oil characterization

In general, pyro oil has more hydrocarbons, and waste plastic pyro oil is similar to fossil fuels like diesel[36]. Plastic oil is recommended as a blending fuel with diesel without any internal combustion engine modifications. Non-condensable gases collected from pyrolysis can also be used as fuels like LPG. Lastly, solid residue in the bottom of the flasks (char) after the pyrolysis reaction can be used as an absorbing material for filtering applications, composite material, and road construction[37].

Density is an important property in fuels because fuel consumption, efficiency, and engine operation depend on density[38]. Diesel fuel has a density of 830 kg/m^3 (Table 4). A higher density results in piston sticking and ignition delay, reducing efficiency. Medical waste plastic oil density was measured as 904 kg/m^3. Viscosity also plays an important role in fuel performance. The viscosity of diesel fuel is commonly around 1.2 cSt, while the viscosity of plastic oil was 2.35 cSt. Flash and fire points are important parameters to gauge fuel performance. Flash and fire points for diesel are around 52 and 56 °C, while the flash and fire points of the medical plastic oil were 42 and 45 °C, respectively. The smoke point is linked with the hydrocarbon composition of the MPO. A high smoke point is favorable for good fuel, and it has a low smoke-producing tendency. The radiant heat transfer of the fuel is more related to the smoke point. The value of the smoke point is 20 mm for medical plastic oil. In engines, fuel injectors, plugs, and small passages are clogged by increasing the fuel's cloud point. The cold operating condition of fuel in the engine fuel injector could lead to wax formation due to low temperature. The cloud point of the MPO is 2 °C. Carbon residue is another problem in fuel, and it raises more problems in engines, such as the absence of ignition, reduced heat transfer rate, and piston sticking, which leads to a reduction in engine operation. The lower the carbon residue, the more valuable the fuel, and it indicates an oil tendency toward coke formation. The carbon residue of the MPO sample is found to be 7.1 wt.%. Cetane number is another important thermos physical property of the fuel, and this number measures the fuel's ignition quality. The cetane number of the MPO is 48, which is quite lower than conventional diesel (52)[39]. As a result of medical plastic oil characterization, it is strongly recommended that it be considered an alternative fuel for IC engines. By-products, such as char and gases, have the potential to be used as absorbent and energy carriers; therefore, the business model of plastic pyrolysis is strengthened and will be key to commercialization soon. Pyrolysis energy consumption and quality can be compromised by the presence of plastic waste pollutants. Table 5 summarizes relevant previous studies and their findings on various aspects of plastic oil production, including the type of feedstock used, the catalysts applied, the processes employed, the resulting yields, and the key inferences drawn regarding plastic oil yield optimization. Also, this highlights the importance of catalyst choice and process conditions in optimizing the yield and quality of the final products, such as oil and gas. For instance, using a zeolite-based catalyst with specific acidity characteristics in a thermocatalytic reactor greatly influences the hydrocarbon fraction, favoring tar elimination and yielding diesel-grade oil. In contrast, when no catalyst is used, the type of plastic heavily dictates the end product composition, with a notable increase in solid residues as the proportion of PET in the feedstock rises. Advanced catalysts, such as MFI-type HZSM-5 zeolite and FCC catalysts, significantly enhance the degree of polymer cracking, leading to higher gas and liquid yields, which are substantially more than yields obtained without catalysts. Moreover, different catalysts and reactor types (e.g., fixed bed, fluidized bed, microwave-assisted) offer varying degrees of effectiveness in breaking down complex polymers, as evidenced by differing yields and product compositions. Overall, the table underscores that the choice of catalyst and pyrolysis method is critical for optimizing outputs, reducing energy requirements, and tailoring the chemical makeup of the resulting pyrolysis products to meet specific industrial needs.

Conclusions

Medical plastic is converted into value-added products using pyrolysis technology, and it is experimentally investigated in a small lab-scale pyrolysis reactor using three different catalysts. Pyrolysis technology is utilized well to reduce the carbon footprint value chain through medical plastic treatment in a close environment. Production of value-added products from waste plastic by catalyst pyrolysis creates awareness about medical plastic waste management. These results conclude catalyst performance depends on the process parameters. Pyrolysis based on the design of the experiment is very useful in identifying the optimized solution to a particular operating condition. Performance optimization of the work is carried out by using DOE software. From the results, the following conclusions have arrived:

- The aluminium oxide catalyst gives more liquid yield than the other two catalysts. The natural zeolite catalyst extracts low liquid yield, but there is a quick start-up during the thermal cracking period, even at low temperatures. On average, natural zeolite produced a 10% higher oil yield than silica oxide catalysts.
- An increase in the reaction temperature using NZ produced a low oil yield compared to Al_2O_3 as a catalyst. This might be due to the higher internal crystal structure of NZ. This is because a high internal crystal structure increases the gas yield at high-temperature regions, consequently reducing the liquid yield.

S.no	Properties	Medical Plastic Oil (MPO)	Plastic oil[10] (PO)	Diesel	Testing standards
1	Density (kg/m^3) (25 °C)	904	906	830	ASTM D 1298
2	Kinematic Viscosity (cSt)	2.35	2.27	1–3.97	ASTM D445 04e
3	Flash & fire point (°C)	42 & 45	42 & 45	52 & 56	ASTM D93
4	Smoke point (mm)	20	20	15	ASTM D1332
5	Cloud point (°C)	2	2	3	ASTM D2500
6	Carbon residue (wt.%)	7.17	7.36	8.2	ASTM D189 05
7	Cetane number	53	52	51	ASTM D 613 05

Table 4. Property comparison of medical plastic oil with standard fuels.

Feedstock	Catalyst	Process	Results / Inference	Reference
HDPE Plastic Pellets (2–4 mm diameter)	Zeolite-based catalyst (Al_2O_3/SiO_2 ratio = 3.5 40–60 50–57 5.5–6), Kaolin clay catalyst	Bench scale thermocatalytic reactor with continuous feed system	Catalyst Lewis acidity key factor in hydrocarbon fraction and higher acidity encourages tar elimination. Extracted oil is warranted as diesel fuel	Auxilio et al. 40
PE rigid/film, PP rigid/film, PET, PS multilayer flexibles and clogged materials	Without catalyst	Laboratory-scale batch reactor (heating rate of 17 °C min to 500 °C)	Plastic type highly decides the end product. Increasing the PET contribution in the feedstock increases solid products.	Genuino et al. 41
HDPE, LDPE, and PP	MFI-type HZSM-5 zeolite (SiO_2/Al_2O_3 =23) and an FAU-type spent fluid catalytic cracking (FCC) catalyst	Fixed bed reactor	Catalysts increase the degree of polymer cracking. Gas (75.2 wt%) and liquid (36%) yields are attained, which are substantially higher than without a catalyst. Catalyst acidity and textural properties highly contribute to liquid yield.	Wong et al. 42
~ 62 wt. %PE (~ 38 wt. % HDPE + ~ 24 wt. % LDPE), polypropylene (~ 34 wt. % PP), polyvinyl chloride (~ 3 wt. % PVC), and with about 1 wt. % polystyrene (PS) mixtures.	FCC catalyst	Fluidized bed reactor	The number of accessible sites along with acid sites is important. FT-IR GCMS are used to predict the compounds and their proportion in the extracted oil.	Lin et al. 43
HDPE, LDPE, PP, PS	A zeolite catalyst loaded with nickel (Ni), molybdenum (Mo), and tungstate (W) referred to as (Z-503)	Fixed bed reactor	67.6 wt. %-89.25 wt.% oil yield obtained by thermal pyrolysis in the order PP > LDPE > HDPE > PS by catalytic pyrolysis. 55.1%-68.2% was attained in the order of LDPE > PP > PS > HDPE by thermal pyrolysis.	Khazaal et al. 44
HDPE and small amount of PP	ZSM -5 catalyst	Continuous microwave-assisted pyrolysis	Oil yield 48.9% and it consists of 73.5% gasoline-based hydrocarbon. Adding ZSM -5 catalyst in the secondary bed further increases the oil yield by 48.9%.	Zhou et al. 45
Post-consumer waste plastics	Zeolite catalysts	Lab-scale pyrolysis reactor	Fresh fluidic catalytic cracking reduce the required activation energy for plastic cracking up to 34% and 6.5% at decomposition's first and second stages.	Kremer et al. 46
Syringes, Medical Bottles	Without catalyst	–	In syringes, thermal cracking occurs at 394.4 to 501 °C, but in medical bottles, this was recorded at 417.9–517 °C.	Ding et al. 47
HDPE and LDPE Medical waste	Natural zeolite Al_2O_3 , SiO_2	Lab scale Pyrolysis Reactor	Aluminum oxide 376 °C and 6.6 wt. % optimist process factors extract 58.3648 and 61.2051 wt. % oil yield. Interaction effect studies depict high acid site catalysts with low temperatures as desirable for both HDPE and LDPE oil yield.	Present study

Table 5. Summary of Various Plastic Pyrolysis Studies.

- The process optimization of DoE is found to be Al_2O_3, temperature 376.6 °C, and Catalyst concentration of 6.5999 is the optimal solution. In this optimum solution, HDPE and LDPE oil extraction are 58.3646% and 61.2051%, respectively.
- The model evaluation results are significant, and the input variables are desirable. The experimental results are authenticated by RSM, along with the R-value above 0.9.
- Medical plastic oil characteristics are tested and compared with commercial fuel. The test results showed that the produced plastic oil exhibits similar properties to commercial fuel and can be used in a diesel engine without any modification.
- By-products char and gases have the potential to be used as absorbents and energy carriers; therefore, the business model of plastic pyrolysis is strengthened and key to commercialization soon. Plastic waste pollutants can compromise pyrolysis energy consumption and quality.
- Overall, this study supports medical plastic waste management in a greener way along with value-added end products and supports the circular economy of plastics.

Data availibility

The datasets used and/or analyzed during the current study are available within the article.

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Author contributions
P.S. R.S. conceived the experiment(s) and conducted the experiment(s), and P.S., U.A., and R.S. analyzed the results. All authors reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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